



# TRANSCRIPT OF RECORDS

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**Hauke Damerow**

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**Faculty of Mathematics, Informatics and  
Natural Sciences**

Desired degree: Master of Science  
Physics

The student has completed the degree.

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Last Name: Damerow

First Name: Hauke

Date of Birth: 19 January 2000

Place of Birth: Bremen

Enrolled on: 1 October 2022

Student ID No.: 7155828

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| Number/Type                               | Name                                                                                                       | Semester/Date | ECTS Credits | Grade       |
|-------------------------------------------|------------------------------------------------------------------------------------------------------------|---------------|--------------|-------------|
| <b>Physics (M.Sc.)</b>                    |                                                                                                            |               | <b>120</b>   | <b>1,44</b> |
| <b>Advanced Master's Courses</b>          |                                                                                                            |               | <b>48</b>    |             |
|                                           | <b>Computational project*</b><br>Recognized achievement (Université de Strasbourg)                         |               | <b>3</b>     | <b>1,0</b>  |
|                                           | <b>Interactions in soft cond*</b><br>Recognized achievement (Université de Strasbourg)                     |               | <b>3</b>     | <b>1,0</b>  |
| <b>PHY-MV-FN-E14</b>                      | <b>Materials research with synchrotron radiation and neutron:<br/>from nanostructures to the component</b> |               | <b>3</b>     | <b>1,0</b>  |
| VL                                        | Materials research with synchrotron radiation and neutrons:<br>From nanostructures to units                | Summer 23     |              |             |
| Final module exam(s):<br>Final exam       |                                                                                                            |               |              | 1,0         |
|                                           | <b>Surfaces and interfaces*</b><br>Recognized achievement (Université de Strasbourg)                       |               | <b>3</b>     | <b>1,3</b>  |
| <b>UE3:</b>                               | <b>Advanced statistical mechanics*</b><br>Recognized achievement (Université de Strasbourg)                |               | <b>6</b>     | <b>1,3</b>  |
| <b>PHY-MV-FN-T35</b>                      | <b>Computational physics with focus on time-dependent<br/>quantum mechanics*</b>                           |               | <b>7</b>     | <b>1,3</b>  |
| PRO                                       | Project in computational physics with focus on time-<br>dependent quantum mechanics*                       | Winter 22/23  |              |             |
| ÜB                                        | Exercises in computational physics with focus on time-<br>dependent quantum mechanics*                     | Winter 22/23  |              |             |
| VL                                        | Computational physics with focus on time-dependent<br>quantum mechanics*                                   | Winter 22/23  |              |             |
| Final module exam(s):<br>Final assignment |                                                                                                            |               |              | 1,3         |
| <b>PHY-MV-LP-T08</b>                      | <b>Quantum technology 2: Atoms and solids</b>                                                              |               | <b>8</b>     | <b>1,0</b>  |
| ÜB                                        | Exercises in quantum technology 2: Atoms and solids                                                        | Summer 23     |              |             |
| VL                                        | Quantum technology 2: Atoms and solids                                                                     | Summer 23     |              |             |
| Final module exam(s):<br>Final exam       |                                                                                                            |               |              | 1,0         |
|                                           | <b>Many body-physics*</b><br>Recognized achievement (Université de Strasbourg)                             |               | <b>3</b>     | <b>1,7</b>  |

| Number/Type                             | Name                                                                                                      | Semester/Date | ECTS Credits | Grade      |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------|---------------|--------------|------------|
| <b>PHY-MV-LP-E29</b>                    | <b>New experiments with XFEL sources*</b>                                                                 |               | <b>4</b>     | <b>1,0</b> |
| ÜB                                      | Exercises in new experiments with XFEL sources*                                                           | Summer 23     |              |            |
| VL                                      | New experiments with XFEL sources*                                                                        | Summer 23     |              |            |
| Exam                                    | Final exam                                                                                                |               |              | 1,0        |
| <b>PHY-MV-LP-E32</b>                    | <b>Seminar about x-ray spectroscopy with FELs and synchrotrons for catalytic and functional materials</b> |               | <b>3</b>     | <b>1,3</b> |
| SEM                                     | Seminar about X-ray spectroscopy with FELs and synchrotrons for catalytic and functional materials        | Winter 22/23  |              |            |
| Final module exam(s):<br>Presentation   |                                                                                                           |               |              | 1,3        |
| <b>PHY-MV-LP-T06</b>                    | <b>Quantum technology 1: Atoms and photons*</b>                                                           |               | <b>8</b>     | <b>1,3</b> |
| ÜB                                      | Exercises in quantum technology 1: Atoms and photons*                                                     | Winter 22/23  |              |            |
| VL                                      | Quantum technology 1: Atoms and photons*                                                                  | Winter 22/23  |              |            |
| Final module exam(s):<br>Block exam     |                                                                                                           |               |              | 1,3        |
| <b>Complementary subject</b>            |                                                                                                           |               | <b>12</b>    |            |
| <b>PHY-CNSM</b>                         | <b>Computer applications in physics: numerical and statistical methods</b>                                |               | <b>4</b>     | <b>b</b>   |
| ÜB                                      | Exercises in computer applications in physics: numerical and statistical methods                          | Summer 23     |              |            |
| VL                                      | Computer applications in physics: numerical and statistical methods                                       | Summer 23     |              |            |
| Final module exam(s):<br>Final exercise |                                                                                                           |               |              | b          |
| <b>Meteorologie KLIMA</b>               | <b>Fundamentals of Meteorology and Climate</b>                                                            |               | <b>4</b>     | <b>b</b>   |
| ÜB                                      | Fundamentals of Meteorology and Climate - Exercise                                                        | Winter 22/23  |              |            |
| VL                                      | Fundamentals of Meteorology and Climate - Lecture                                                         | Winter 22/23  |              |            |
| Final module exam(s):<br>Final Exercise |                                                                                                           |               |              | 1,3        |
| <b>MET-M-RC</b>                         | <b>Radiation and Climate*</b>                                                                             |               | <b>6</b>     | <b>b</b>   |
| ÜB                                      | Exercises in Radiation and Climate*                                                                       | Winter 24/25  |              |            |
| VL                                      | Radiation and Climate, Lecture*                                                                           | Winter 24/25  |              |            |
| Final module exam(s):<br>Presentation   |                                                                                                           |               |              | 2,3        |

| Number/Type                                                                                                                              | Name                                                                                      | Semester/Date    | ECTS Credits | Grade      |
|------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|------------------|--------------|------------|
| <b>Research Phase</b>                                                                                                                    |                                                                                           |                  | <b>30</b>    |            |
| <b>PHY-MF-LP-EP</b>                                                                                                                      | <b>Introductory project (Laser physics and photonics)</b>                                 |                  | <b>15</b>    | <b>b</b>   |
| Final module exam(s):                                                                                                                    |                                                                                           |                  |              |            |
| Final assignment                                                                                                                         |                                                                                           |                  |              | b          |
| <b>PHY-MF-LP-VPb</b>                                                                                                                     | <b>Preparatory project (Laser physics and photonics)</b>                                  |                  | <b>15</b>    | <b>b</b>   |
| Final module exam(s):                                                                                                                    |                                                                                           |                  |              |            |
| Final assignment                                                                                                                         |                                                                                           |                  |              | b          |
| <b>Master thesis</b>                                                                                                                     |                                                                                           |                  | <b>30</b>    |            |
| <b>PHY-MF-MA</b>                                                                                                                         | <b>Master's thesis</b>                                                                    |                  | <b>30</b>    | <b>1,7</b> |
| Final Paper                                                                                                                              |                                                                                           | 29 Sep 2025      |              | 1,7        |
| Master's thesis: Fast absorption parameterization and radiative flux computation with the atmospheric radiative transfer simulator ARTS* |                                                                                           |                  |              |            |
| Examiners: Prof. Dr. Peter Hauschildt, Prof. Dr. Stefan Alexander Buehler                                                                |                                                                                           |                  |              |            |
| Oral exam                                                                                                                                |                                                                                           | 2 Oct 2025       |              | 2,0        |
| <b>Additional</b>                                                                                                                        |                                                                                           |                  | <b>0</b>     |            |
| <b>UE1:</b>                                                                                                                              | <b>Advanced quantum mechanics*</b><br>Recognized achievement (Université de Strasbourg)   |                  | <b>6</b>     | <b>2,7</b> |
| <b>PHY-MV-FN-E01</b>                                                                                                                     | <b>Advanced solid-state physics</b>                                                       |                  | <b>8</b>     | <b>2,3</b> |
| ÜB                                                                                                                                       | Excercises in advanced solid-state physics                                                | Summer 23        |              |            |
| VL                                                                                                                                       | Advanced solid-state physics                                                              | Summer 23        |              |            |
| Final module exam(s):                                                                                                                    |                                                                                           |                  |              |            |
| Written exam                                                                                                                             |                                                                                           | 24 Jul 2023      |              | 2,3        |
| <b>VL + ÜB</b>                                                                                                                           | <b>Integrated Climate-Economic Modeling</b>                                               | <b>Summer 25</b> | <b>3</b>     | <b>b</b>   |
| Exam                                                                                                                                     | Block exam                                                                                |                  |              | b          |
| <b>ESW-001</b>                                                                                                                           | <b>Sustainability: Concepts, Examples, and Shaping Our Future</b>                         |                  | <b>3</b>     | <b>b</b>   |
| VL                                                                                                                                       | Sustainability: Concepts, Examples, and Shaping Our Future                                | Winter 24/25     |              |            |
| Final module exam(s):                                                                                                                    |                                                                                           |                  |              |            |
| Block exam                                                                                                                               |                                                                                           | 13 Feb 2025      |              | 1,0        |
| <b>UE2:</b>                                                                                                                              | <b>Radiation matter interaction*</b><br>Recognized achievement (Université de Strasbourg) |                  | <b>6</b>     | <b>2,7</b> |

| Number/Type          | Name                                                                                                                                                           | Semester/Date    | ECTS Credits | Grade       |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|--------------|-------------|
| RVL                  | Lecture Series on Climate Crisis: Another World Is Possible!                                                                                                   | Winter 22/23     | 4            | b           |
| Exam                 | Online tests                                                                                                                                                   |                  |              | b           |
| RVL                  | Lecture Series on Climate Crisis: Hot Planet, Cold Facts                                                                                                       | Winter 24/25     | 3            | b           |
| Exam                 | Online tests                                                                                                                                                   |                  |              | b           |
|                      | Scattering techniques*<br>Recognized achievement (Université de Strasbourg)                                                                                    |                  | 3            | 2,3         |
| PHY-MV-FN-T18        | Seminar on many body systems and quantum statistical methods*                                                                                                  |                  | 3            | 1,7         |
| SEM                  | Seminar on many body systems and quantum statistical methods*                                                                                                  | Summer 23        |              |             |
| Exam                 | Presentation and paper                                                                                                                                         |                  |              | 1,7         |
| ESW-002              | Cutting-edge Research for Sustainability in Hamburg - Contributions of the Faculty of Mathematics, Informatics and Natural Sciences to Sustainable Development |                  | 3            | b           |
| VL                   | Cutting-edge Research for Sustainability in Hamburg - Contributions of the Faculty of Mathematics, Informatics and Natural Sciences to Sustainable Development | Summer 25        |              |             |
| Exam                 | Block exam                                                                                                                                                     |                  |              | b           |
| <b>Current grade</b> |                                                                                                                                                                | <b>excellent</b> |              | <b>1,44</b> |

A minimum of 120 ECTS credits is required to successfully complete the program.

Hamburg, 28 November 2025

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Note: This document includes only successfully completed courses and examinations.

**Grading system per component:**

|                 |   |              |
|-----------------|---|--------------|
| 1,0 / 1,3       | = | excellent    |
| 1,7 / 2,0 / 2,3 | = | good         |
| 2,7 / 3,0 / 3,3 | = | satisfactory |
| 3,7 / 4,0       | = | sufficient   |
| 5,0             | = | insufficient |

Calculations for the final grade and for individual components are determined by departmental regulations.

\* = in the original language

b = pass

inc. = module/course incomplete

e = successfully completed